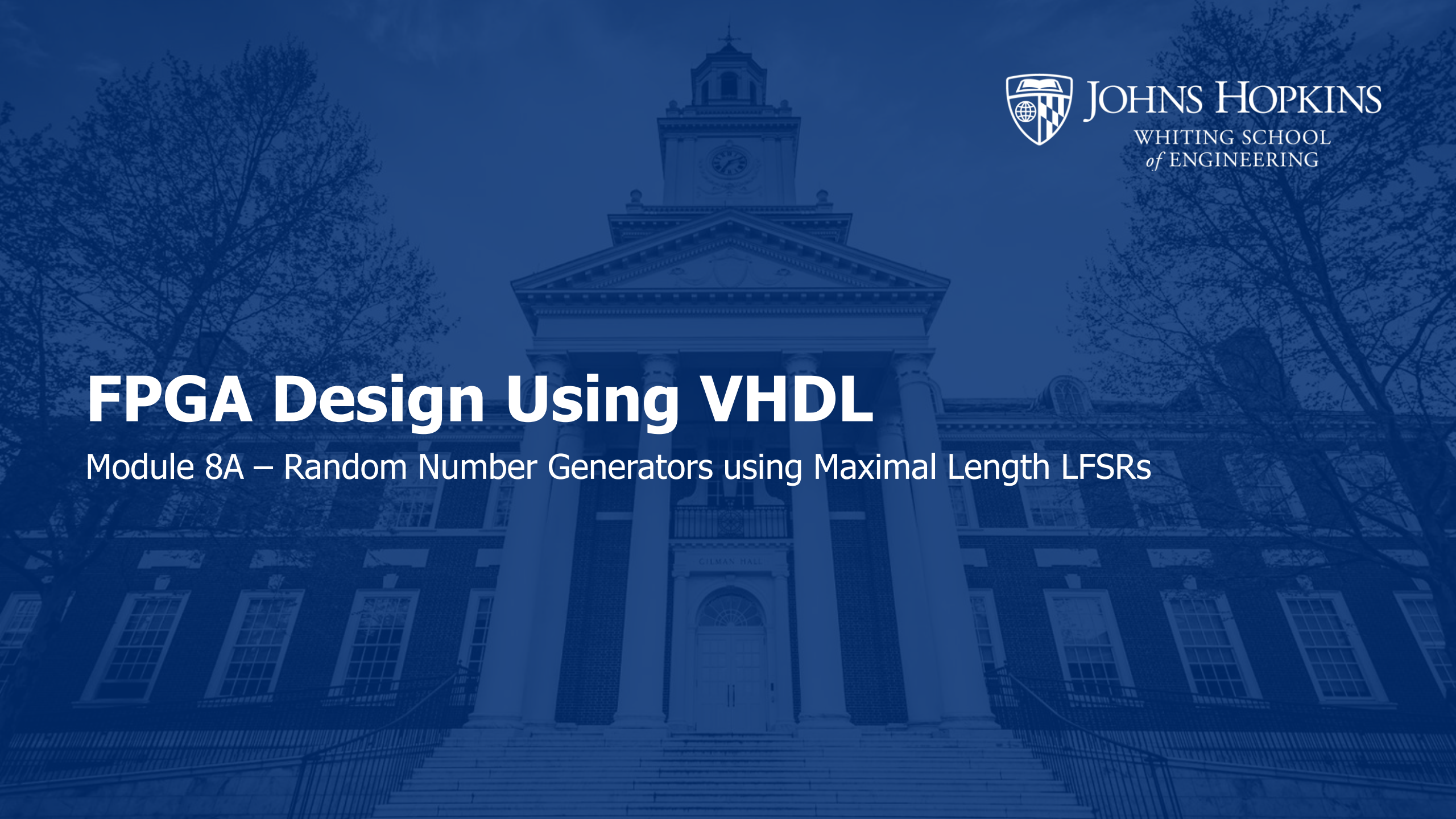


FPGA Design Using VHDL

Module 8A – Random Number Generators using Maximal Length LFSRs

In this Module, You Should …

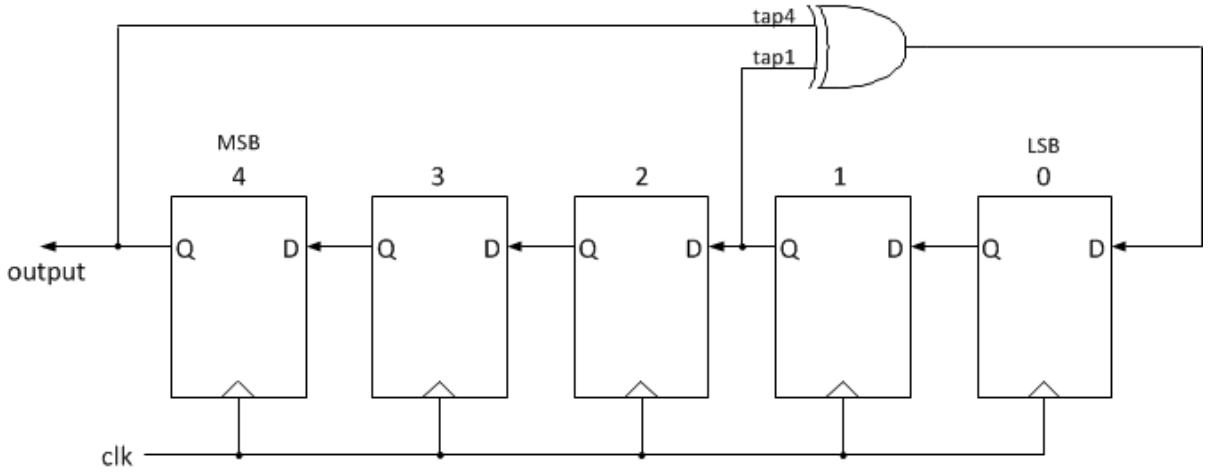
- Describe a Linear Feedback Shift Registers (LFSR)
- Differentiate between purely random and pseudo-random.
- Create a pseudo-random number generator using LFSR.
- Recognize what is a maximal length LFSR.
- Design an LFSR using VHDL.

Pseudo Random Number Generators

- **Genuinely random** numbers: knowledge of one number or arbitrarily long sequence of numbers is no use in predicting the next number to be generated
- **Pseudo-random**: You can predict all numbers given knowledge of algorithm used to create the numbers and its initial state
- **Computer-generated** "random" numbers
  - Properly referred to as pseudo-random numbers,
  - They exhibit characteristics of genuinely random numbers

Linear Feedback Shift Register

- LFSRs are *n* shift registers
  - The sequence of valid states always repeat
  - Some states may be skipped (will discuss why later)
  - Value of states exhibit pseudo-random behavior
- Designed using simple shift-registers with a small number of `xor` gates for the feedback

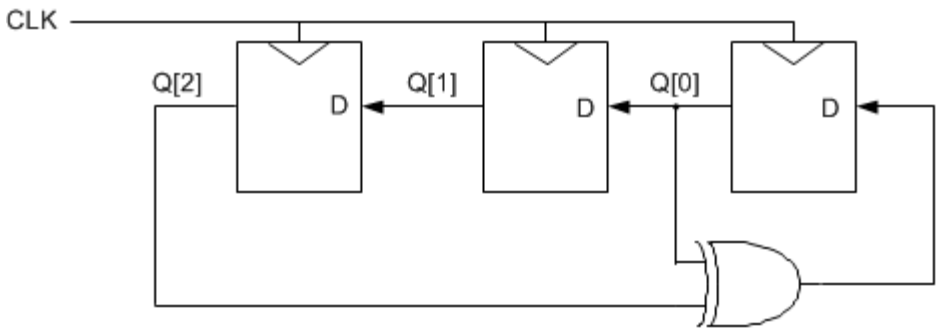


Linear Feedback Shift Register Continued

- **Maximum** number of possible states: 2^n-1
  - Lockup state: A state that you cannot recover from
    - All zeros state
  - Your seed (initial state) should NEVER be all zero
- Always best to design "un-lock" circuitry
  - Timers, resets, etc…
  - It is possible in radiation environments that a single bit can flip from 1 to 0 causing the potential all zeros state

Linear Feedback Shift Register Continued More

- Assume a 3-bit LFSR with initial seed "111"
- Outputs of DFF0 and DFF2 are XOR-ed and fed back to input



Step	Q2 Q1 Q0
0	111
1	110
2	101
3	010
4	100
5	001
6	011
7	111
8	Sequence Repeats

Maximal Length LFSR

- Certain "taps" from the shift registers can be selected for the feedback to ensure each sequence (state) is encountered at most once during 2^n-1 cycles
- Taps are usually represented as polynomials:
  - $x^{n-1} + x^{n-2} + \ldots + x + 1$
  - "+" represents modulo 2 adder (xor)
  - Polynomial from previous example: $x^2 + 1$
  - Coefficients are either '0' or '1'
  - Coefficient for x^{n-1} always '1' for maximal length
- Maximal Length LFSRs should satisfy the following properties:
  - Have a tap on the last flip flop in the LFSR
  - For any non-zero seed, the LFSR has (2^n)-1 cycles prior to repeating
- In this course, the best method to test for maximal length is to start with any non-zero seed value on the LFSR registers and calculate each iteration until you get a repeated value

Maximal Length LFSR Continued

- Table of Maximal Length Polynomials:
  - http://www.ece.cmu.edu/~koopman/lfsr/index.html

